

Resubmission:
No

Questions And Suggestions For Rebuttal:

1. The author claims to consider 'all possible time structure' via the time integral in Eq (2). However, this relies heavily on Assumption 1 (that the training set is representative enough). If the test set exhibits a temporal structure f_{test} that is outside the support of $p(f)$, the 'Time-Invariant' property might fail. Does the author provide an analysis on how much data is 'sufficient' to represent 'all possible structures'?
2. On the ETTm dataset, the improvement brought by S compared to random initialization of weights is modest, and it also increased the standard deviation, which may introduce additional instability into the training process, making the model less robust than simple random initialization. The authors should explain why a principled starting point leads to higher variance and whether this trade-off is acceptable in real-world deployments.
3. There are multiple broken cross-references to critical evidence (e.g., Table T1 in "Frequency modeling ablation studies", Appendix B.1, and Appendix B.3). These missing elements prevent a thorough evaluation of the proposed ablation studies and efficiency claims.
4. The authors state that the best results are highlighted in bold. However, in Table 16, the MSE for eqn. 8.9 (0.3932) is better than the bolded value for no_update (0.3938). Such inconsistent and incorrect formatting further diminishes the credibility of the reported experimental analysis. (There are several grammatical issues throughout the paper for example, "... so it can sufficiently representation the distributions". A careful proofreading would improve the overall clarity of the manuscript.

Relevance: 3: Moderate - The work is somewhat relevant to the Research track of KDD and is of narrow interest to a sub-community

Novelty: 2: Low - The ideas are relatively minor and largely incremental. The work builds heavily on existing research.

Technical Quality: 3: Moderate - The paper demonstrates solid technical quality with a sound methodology and thorough analysis. The results are reliable and well-supported. There may be minor issues, but they do not significantly undermine the overall quality. The work is competently executed and meets acceptable standards.

Presentation: 1: Fair - The paper is poorly organized and difficult to follow. The writing is riddled with numerous grammatical and typographical errors. Figures and tables, if present, are poorly designed or hard to understand. Overall, the presentation detracts significantly from the readability and comprehension of the work.

Reproducibility: 2: Low - The paper includes some information about the methods, data, and procedures, but key details are missing. There may be supplementary materials, but they are incomplete or unclear. Reproducing the results would require significant effort and additional information.

Reviewer Confidence: 3: Moderate - The reviewer has a good understanding of the subject area and is familiar with the research methods and context of the paper. They feel confident in their ability to accurately assess the quality and significance of the work. Their evaluation is based on a solid grasp of the content and context.

Ethics Review Flag: No

Ethics Review Description: No

Official Review of Submission991 by Reviewer 54Tf

Official Review: by Reviewer 54Tf 20 Mar 2026, 15:35 (modified: 16 May 2026, 11:29) Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer 54Tf, Authors

Revisions

Paper Summary:
This paper proposes the time-invariant frequency operator to address non-stationarity in time series forecasting from a frequency-domain perspective. TIFO leverages global dataset statistics (the signal-to-noise ratio (mean divided by variance of frequency amplitudes) calculated offline). During training, these stationary scores act as a fixed prior for an MLP-based modulation layer that adaptively weights frequency components to highlight stable patterns and suppress non-stationary ones. TIFO is designed as a plug-and-play module and is evaluated across multiple backbones (DLinear, PatchTST, Transformer). Experimental results on non-stationary datasets like ETTm show significant improvements in MSE and MAE compared to existing normalization baselines.

Paper Strengths:

1. This paper is well-written, with clear logic and a well-organized structure. The author provides a workflow for the algorithm and clearly explains its concepts and principles.
2. It proposes to use dataset-wide statistics (mean/variance of amplitudes) as a stationary score to guide sample-level modulation.
3. It demonstrates seamless integration with diverse architectures, including linear models and Transformers, and it can be easily integrated into multiple backbones.

Paper Weaknesses:

1. According to Figure 1, Step 1 to Step 2 extracts and separates frequency features f_k into stationary (f_s) and non-stationary (f_n) components. However, Section 3.1 describes a different mechanism where the MLP takes pre-calculated scores S as input to produce a single weight λ for modulation. There is no separation operation as depicted in the diagram. This creates a severe disconnect between the visual representation of the method and its actual implementation.
2. The dimension of the amplitude matrix A is not specified.
3. Sections 2.1, 2.2, and 2.3 contain overlapping discussions regarding distribution shifts and the $p(f)$ problem.

Resubmission:
N/A

Questions And Suggestions For Rebuttal:
The proposed mechanism is somewhat lightweight and simple. What's the potential of the proposed method to work as a general feature enhancement tool beyond long-term forecasting, such as imputation and classification?

Relevance: 3: Moderate - The work is somewhat relevant to the Research track of KDD and is of narrow interest to a sub-community

Novelty: 2: Low - The ideas are relatively minor and largely incremental. The work builds heavily on existing research.

Technical Quality: 3: Moderate - The paper demonstrates solid technical quality with a sound methodology and thorough analysis. The results are reliable and well-supported. There may be minor issues, but they do not significantly undermine the overall quality. The work is competently executed and meets acceptable standards.

Presentation: 3: Moderate - The paper is organized and generally clear. The writing is mostly free of grammatical and typographical errors, making it easy to read. Figures and tables are effectively used to support the text. The presentation facilitates understanding and conveys the key points effectively.

Reproducibility: 3: Moderate - The paper provides a clear and detailed description of the methods, data, and procedures used. Supplementary materials, such as datasets and code, are available and sufficiently documented. Reproducing the results would be feasible with the provided information, though some effort may still be required.

Reviewer Confidence: 4: High - The reviewer is an expert in the subject area and has extensive knowledge of the research methods and context of the paper. They are highly confident in their ability to provide an accurate and thorough assessment. Their evaluation is based on deep expertise and a comprehensive understanding of the work.

Ethics Review Flag: No

Ethics Review Description: no

Official Review of Submission991 by Reviewer e8mV

Official Review: by Reviewer e8mV 16 Mar 2026, 11:32 (modified: 16 May 2026, 11:25) Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer e8mV, Authors

Revisions

Paper Summary:
The authors proposed a Time-Invariant Frequency Operator (TIFO), which learns stationarity-aware weights over the frequency spectrum across the entire dataset. The weight representation highlights stationary frequency components while suppressing non-stationary ones, thereby mitigating the distribution shift issue in time series.

Paper Strengths:
The stationarity-aware weight representation highlights stationary frequency components while suppressing non-stationary ones, thereby mitigating the distribution shift issue in time series. TIFO is a plug-and-play approach that can be seamlessly integrated into various forecasting models.

Paper Weaknesses:

1. From normalization methods to TIFO, there are no evidence to support their relationship. What is the relationship between Normalization methods and Time-Invariant Frequency Operator?
2. Why do frequency shifts could replace the distributional shift of training datasets and test datasets? Maybe it is only one part of the distributional shift of datasets.
3. How could TIFO could solve the distributional shift with the theoretical analysis? It lacks theoretical justification.
4. The baseline methods are weak. It lacks SOTA algorithms from 2025 and 2026.
5. How could TIFO obtains stationarity-aware representation in the experiment? Which experimental setting could support the statement?

Resubmission:
N/A

Questions And Suggestions For Rebuttal:

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Relevance: 3: Moderate - The work is somewhat relevant to the Research track of KDD and is of narrow interest to a sub-community

Novelty: 3: Moderate - The paper introduces a new and interesting idea or approach that adds value to the field. The contribution is original and represents an advancement of existing knowledge, demonstrating solid innovation and creativity.

Technical Quality: 3: Moderate - The paper demonstrates solid technical quality with a sound methodology and thorough analysis. The results are reliable and well-supported. There may be minor issues, but they do not significantly undermine the overall quality. The work is competently executed and meets acceptable standards.

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Ethics Review Flag: No

Ethics Review Description: N/A

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